

WASP-36b

R-1912

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_160216 · created 2026-08-05T23:44:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457434.6595 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.049
Transit depth (Rp/Rs)^2	2.77 +/- 1.62 [%]
Orbital Inclination (inc)	84.79 +/- 2.83
Airmass coefficient 1 (a1)	0.63 +/- 0.36
Airmass coefficient 2 (a2)	0.32 +/- 0.17
Scatter in the residuals of the lightcurve fit is	53.96 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	4.75
Transit Duration (day)	0.082 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.049 vs 0.1368 (deviation 0.6σ)
Duration fit vs published	0.082 d vs 0.0758 d (deviation 0.33σ)
Mid-time O–C	-4.2 min
Depth SNR	3.3
Residual scatter	53.96 %

Light curve

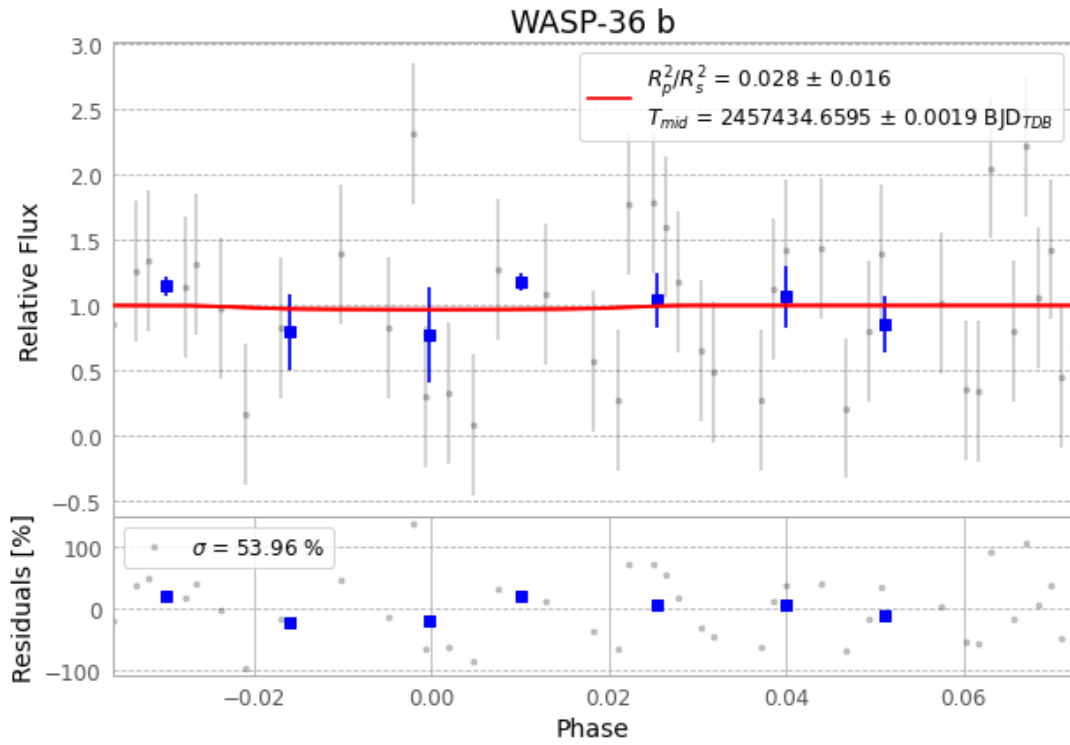


Figure 1 — FinalLightCurve_WASP-36 b_2016-02-15.png (SHA-256 c0a6432317f4ea84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 360], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 225cb59a568ed87b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), WASP-36160216022112.FITS → WASP-36160216062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-160216.log (b8d07b52f0b2...), FinalLightCurve_WASP-36 b_2016-02-15.png (c0a6432317f4...), FinalLightCurve_WASP-36 b_2016-02-15.csv (685f4e629e2e...)

Compute resources

Wall time 187.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:44Z.